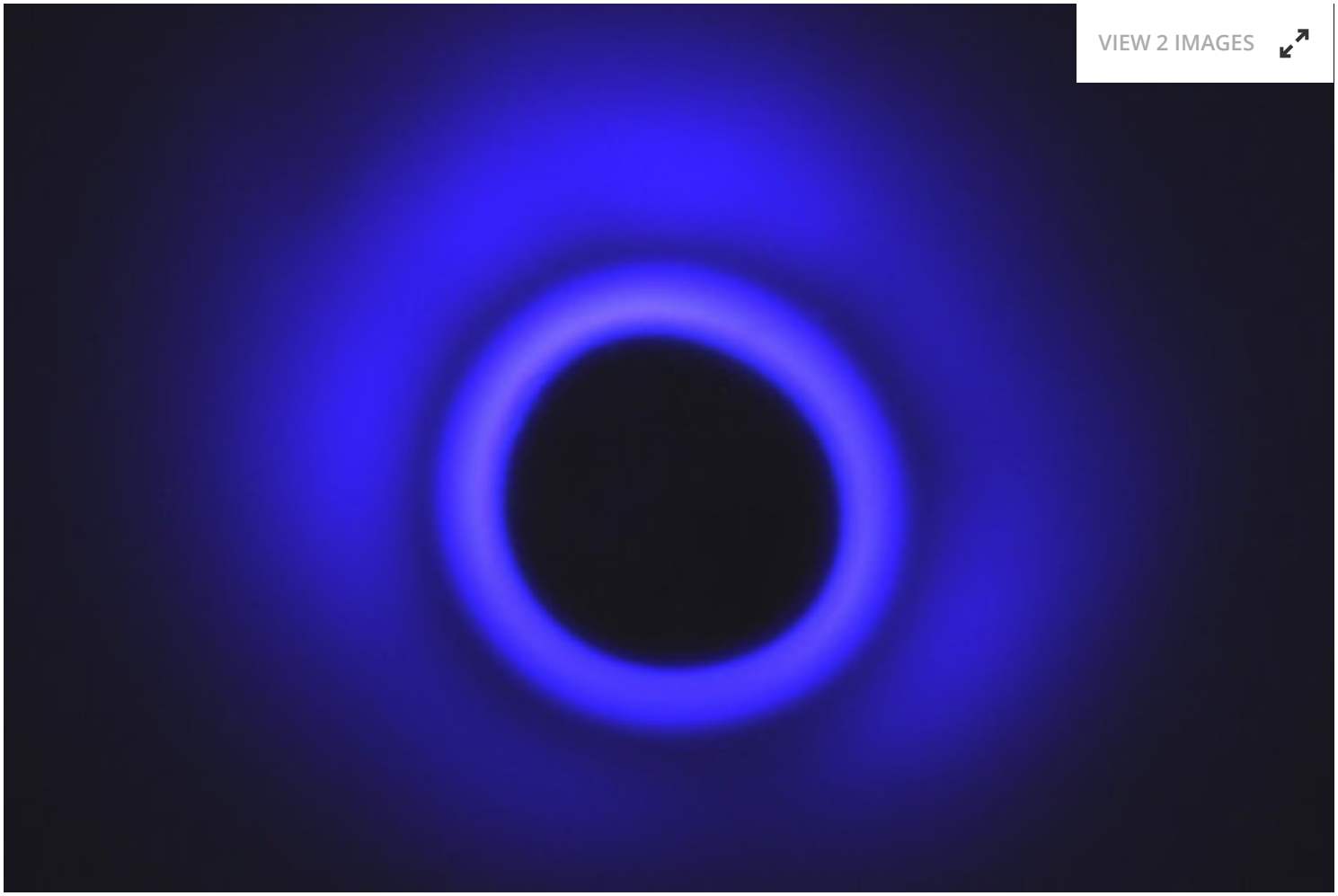
PHYSICS

Stable plasma ring created in open air for the first time



Caltech engineers have created a stable ring of plasma in the open air **Mory Gharib/Caltech**

Plasma is one of the more mysterious phases of matter, only existing on Earth fleetingly in the form of lightning or contained inside neon signs. Now a team of engineers from Caltech has created a stable ring of plasma in open air, using just a high-pressure jet of water and a crystal

plate. The technique could, in the long run, unlock the potential for plasma to be used for storing energy.

Plasma is created when an ionized gas is electrically charged, and being gaseous, it normally doesn't have a set shape. When made artificially, it's usually constrained in vacuum chambers or electromagnetic fields, but previous work has generated plasma rings in the open air for up to 10 milliseconds at a time. The Caltech team managed to create rings that lasted indefinitely, as long as a power source is maintained.

"We were told by some colleagues this wasn't even possible," says Francisco Pereira, co-author of the study. "But we can create a stable ring and maintain it for as long as we want, no vacuum or magnetic field or anything."

To make these plasma rings, the engineers simply squirt water at a crystal plate. That jet of water is extremely narrow and highly pressurized, measuring just 85 microns wide and with a force of 9,000 psi, which strikes the plate with an impact velocity of about 1,000 ft per second (305 m per second). To illustrate those numbers, the team likens the jet to a human hair moving at the speed of a bullet.

Stable Plasma Torus

The crystal surface is negatively charged, and when the water hits it and flows outwards, it carries with it positively-charged ions. That builds up a static (or triboelectric) charge, which causes electrons to flow up to the surface of the water and ionize the atoms and molecules in the air above it. That creates a ring of plasma in the air, which will stay steady as long as the stream of water is flowing.

The rings have a diameter of a few dozen microns and are visible under a microscope. Strangely, the plasma appears to be emitting radio frequencies, which messed with the engineer's cell phones while they were in the room.

"That's never been seen before," says Pereira. "We think it's because of the piezo properties (an electric charge generated through mechanical stress) of the materials that we used in our experiments."

While the technique is more curiosity than commercial, the team says that generating plasma without needing powerful electromagnetic fields or vacuums could make plasma-based energy storage systems possible in the future.

The research will be published this week in the *Proceedings of the National Academy of Sciences*.

Source: Caltech

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Michael has always been fascinated by space, technology, dinosaurs, and the weirder mysteries the universe. With a Bachelor of Arts in Professional Writing and several years experience under his belt, he joined New Atlas as a staff writer in 2016.

5 COMMENTS

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JohnHL NOVEMBER 14, 2017 08:17 AM

Yet another "could be" energy solution!

noteugene NOVEMBER 14, 2017 02:12 PM

I'm not an engineer so I don't understand how this could ever be used an an energy source or a method for storing energy. Would'nt more energy be used in blasting water than the small amount of plasma generated? That's not to say that this isn't an interesting discovery. I could use enlightment.

Dan Lewis NOVEMBER 14, 2017 07:56 PM

Ooooh! Pretty! ;)

HoppyHopkins NOVEMBER 15, 2017 02:32 AM

So it can be used to store energy. I can see plasma guns in the near future for the Military and the Elmer Fud types to play with

Saigvre DECEMBER 12, 2017 04:00 AM

The way it's written up, it belongs in PNAS, but it could also show in r/powerwasherporn (reddit!) See? http://www.pnas.org/content/114/48/12657/F1.expansion.html Shear waves radiating from the strike area...aw, see figure 6. If you're doing the single tiny laminar-flow orifice thing like they are, you need almost 0 flow (tiny jet of DI water, bubbles all out) at 9000 psi, the camera behind the Quartz or LiBeO3 target with a microscope racked up on it, and I guess someone to pull focus back there. Setup's in the supplemental info of the whole Open Access thing: http://www.pnas.org/content/114/48/12657.full.pdf?with-ds=yes Put failsafes on fingers (siblings' iPads, etc.) meeting the water jet (and on bubbles in the water,) and you've got a plasma physics setup you don't have to file with the DOE about. Unless you borrow each others' power washers. Thanks, CalTech and National Academy of Science (Proceedings!)

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